

Def.

Let G be a group and A be a non-empty subset. A map

$$\alpha: G \times A \rightarrow A: (g, a) \mapsto g \cdot a$$

is called a group action of G on A if it satisfies

- 1) For all $a \in A$, $1_G \cdot a = a$
- 2) For all $g_1, g_2 \in G$, and $a \in A$, $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

Prop. Given a group action α of G on A , there exists a homomorphism

$$\mathcal{Q}_\alpha: G \rightarrow S_A: g \mapsto \sigma_g \quad \text{where} \quad \sigma_g: A \rightarrow A: a \mapsto g \cdot a.$$

which is called the permutation representation associated to the given action.

Conversely, given a homomorphism $\mathcal{Q}: G \rightarrow S_A$, there exists a group action

$$\alpha_{\mathcal{Q}}: G \times A \rightarrow A: (g, a) \mapsto \mathcal{Q}(g)(a)$$

Proof. We only need to check that the given map is well-defined,

Let $g \in G$ be fixed. Then, $\sigma_g \circ \sigma_{g^{-1}}(a) = \sigma_{g^{-1}} \circ \sigma_g(a) = a$, therefore $\sigma_g \in S_A$.

The converse is obtained by the definition of group action.

Note: The concept of group action is: for a set A , determine a permutation

$$A \rightarrow A: a \mapsto g \cdot a$$

from each element $g \in G$, compatible with the group structure of G , in the sense: the afforded homomorphism $\mathcal{Q}_\alpha: G \rightarrow S_A$ preserves the operation.

Def. Let G acts on A . For each $a \in A$, a subset

$$\text{Orb}_G(a) \stackrel{\text{def}}{=} \{g \cdot a \mid g \in G\} \subseteq A$$

is called the Orbit of a .

Prop. Let a group G acts on a set A . Given $a \in A$, a map

$$G/G_a \rightarrow \text{Orb}_G(a): gG_a \mapsto g \cdot a$$

is bijection. In particular, if G is finite, then

$$|G:G_a| = \#\text{Orb}_G(a).$$

Proof. Given $g_1, g_2 \in G$,

$$g_1 G_a = g_2 G_a \Leftrightarrow g_1^{-1} g_2 \in G_a \Leftrightarrow g_1^{-1} g_2 a = a \Leftrightarrow g_1 a = g_2 a$$

So it is well-defined and injective. And the surjectiveness is clear.

Prop. (Dummit, 4.1.9). Let G acts transitively on the finite set A , and $|A| \leq |G|$ be normal. Let $\mathcal{O}_1, \dots, \mathcal{O}_r$ be the distinct orbits of H on A

a). For each $g \in G$ and $i \in \{1, \dots, r\}$, there is a j such that $g\mathcal{O}_i = \mathcal{O}_j$.

Further, G is transitive on $\{\mathcal{O}_1, \dots, \mathcal{O}_r\}$, therefore all $\mathcal{O}_i$ have same cardinality.

b). If $a \in \mathcal{O}_1$, then $|\mathcal{O}_1| = |H:H \cap G_a|$ and therefore $r = |G:H \cap G_a|$.

Proof. Let $x \in \mathcal{O}_i$. Then, we can write $\mathcal{O}_i = \{hx_i \mid h \in H\}$. Given $g \in G$,

$$g\mathcal{O}_i = \{gx_i \mid h \in H\} = \{h'gx_i \mid h' = ghg^{-1} \in H\} = \text{Orb}_H(gx_i) = \mathcal{O}_j, \text{ for some } j.$$

Let $\mathcal{O}_1$ be fixed. Then, for any $\mathcal{O}_j$ with representative x_j , there is $g \in G$ such that $gx_1 = x_j$, so $g\mathcal{O}_1 = \mathcal{O}_j$, it proves a).

By the second-isomorphism theorem, given $a \in \mathcal{O}_1$,

$$H G_a / H \cong G_a / H \cap G_a$$

$$\begin{array}{c} G \\ \downarrow \\ H G_a \\ \swarrow \quad \searrow \\ H \quad G_a \\ \downarrow \quad \downarrow \\ H \cap G_a \end{array}$$

so, $|\mathcal{O}_1| = |H:H \cap G_a|$ and $|A| = r \cdot |\mathcal{O}_1|$ implies

$$|A| = |G:G_a| = |G:H \cap G_a| \cdot |H G_a:G_a| = |G:H \cap G_a| \cdot |H:H \cap G_a| = |G:H \cap G_a| \cdot |\mathcal{O}_1|.$$

gives the result.

Thm.

Let G be a group and $H \leq G$ be a subgroup.

Consider G acts on $\{gH \mid g \in G\}$ by left multiplication, and affords homomorphism

$$\pi_H: G \rightarrow S_{G/H}: g \mapsto \sigma_g \text{ where } \sigma_g(xH) = gxH.$$

Then G acts transitively on A and the stabilizer $G_{1 \cdot H} = H$, and

$$\ker \pi_H = \bigcap_{x \in G} xHx^{-1} \leq H$$

is the largest normal subgroup of G contained in H .

Proof. Given $aH, bH \in A$, $bH = ba^{-1}(aH)$, so aH and bH lie in same orbit.

And, since $gH = H \iff g \in H$,

$$G_{1 \cdot H} = \{g \in G \mid gH = H\} = H.$$

To show the last assertion, by definition,

$$\ker \pi_H = \bigcap_{x \in G} G_{xH}$$

and $G_{xH} = \{g \in G \mid gxH = xH\} = \{g \in G \mid x^{-1}gx \in H\} = xHx^{-1}$.

Moreover, $\ker \pi_H \leq H$ because it is kernel of the homomorphism, and for any normal subgroup $N \leq G$ of G contained in H ,

$$\text{for all } x \in G, N = xNx^{-1} \leq xHx^{-1}$$

therefore $N \leq \ker \pi_H$. $\square$

Prop. If $H \leq G$ has finite index $n = |G:H|$, then there is a normal subgroup $K \leq G$ s.t.

$$K \leq H \text{ and } |G:K| \mid n!$$

Proof. The kernel of $\pi_H: G \rightarrow S_{G/H}: g \mapsto (xH \mapsto gxH)$ is contained in H . And $G/\ker \pi_H \cong \text{Im } \pi_H \leq S_{G/H} \cong S_n$. Thus $|G:\ker \pi_H| \mid n!$, and

$$|G:\ker \pi_H| = |G:H| \cdot |H:\ker \pi_H| = n \cdot |H:\ker \pi_H|$$

so $|H:\ker \pi_H|$ divides $(n-1)!$.

Thm.

Let G be a finite group of order n , and let p be the smallest prime dividing n . Then, every subgroup of index p is normal.

Proof. Let $H \leq G$ with $|G:H| = p$. Let G acts on left cosets of H and

$$\pi_H: G \rightarrow S_{G/H}$$

be the afforded permutation of the given action.

Then, $\ker \pi_H \leq H$, let $K = \ker \pi_H$ and $p = |H:K|$. Then,

$$|G:K| = |G:H| \cdot |H:K| = p^2.$$

Since $G/K \cong \text{Im } \pi_H \leq S_p$, so p^2 divides $p!$ by the Lagrange's Theorem.

Therefore, p divides $(p-1)!$. But all prime factors of $(p-1)!$ are less than p , by the minimality of p , p must be 1, that is, $H = K = \ker \pi_H \leq G$ and

Note: let G acts on finite set A which has m elements.

Then, the afforded permutation $\varphi: G \rightarrow S_m$ satisfies:

$$G/\ker \varphi \cong \text{Im } \varphi \leq S_m \text{ implies } |\text{Im } \varphi| \mid |G| \text{ and } |\text{Im } \varphi| \mid m!$$

So, if p is the smallest prime factor of $|G|$, then

$$m < p \Rightarrow \text{Im } \varphi = 1 \quad \text{and} \quad m = p \Rightarrow |\text{Im } \varphi| \in \{1, p\}.$$

In the case of the above theorem, the action is transitive, therefore $\text{Im } \pi_H \neq 1$. Therefore, $|\ker \pi_H| = |H|$ and $\ker \pi_H \leq H$, so proved.

Def. Let G be a group, and let $H \leq G$ be a subgroup.

A subgroup $\text{Core}_G(H) \stackrel{\text{def}}{=} \bigcap_{g \in G} gHg^{-1}$ is called the core of H in G . (It is normal).

Note: the theorem above gives

$$H \leq G \Leftrightarrow H = \text{Core}_G(H).$$

Problems in dummit, 4.2.

Thm. If G is a group of order p^n where p is prime and $n \in \mathbb{N}$, then every subgroup of index p is normal.

Proof. Let $H \leq G$ be the subgroup with $|G:H| = p$.

Then, the Homomorphism defined by the left Coset action

$$\varphi: G \rightarrow S_{G/H}; g \mapsto (xH \mapsto gxH)$$

satisfies

$$G/\ker \varphi \cong \text{Im } \varphi \leq S_{G/H}$$

So, $|G/\ker \varphi|$ divides $p!$, but since p is prime, $|G/\ker \varphi| \in \{1, p\}$. Since $\ker \varphi \leq H \neq G$, $\ker \varphi \neq G$, so $|G/\ker \varphi| = p$, that is, $|\ker \varphi| = p^{n-1}$. Consequently, $H = \ker \varphi \trianglelefteq G$ is normal. $\square$

Notes: If G is group of order p^2 , then it has order p element $x \in G$ by Cauchy's theorem, and $\langle x \rangle \trianglelefteq G$.

Thm. Group of order 6 are only $\mathbb{Z}/6\mathbb{Z}$ and S_3 .

Proof. If G is abelian, then the Cauchy theorem gives order 2 element x and order 3 element y . Since x and y commute, $|xy| = 6$, so $G = \langle xy \rangle \cong C_6$. Assume that G is non-abelian. By the Cauchy theorem again, there exists order 2 element x . Let $H = \langle x \rangle$. Then, the homomorphism

$$\tau_H: G \rightarrow S_{G/H}; g \mapsto (xH \mapsto gxH)$$

satisfies that $G/\ker \tau_H \cong \text{Im } \tau_H \leq S_{G/H} \cong S_3$.

If H is normal, then for all $g \in G$, $gxg^{-1} \in \langle x \rangle$, that is, $gxg^{-1} = x$, so x commutes with order 3 element y , it implies $\langle xy \rangle = 6$, it contradicts. Therefore H is non-normal, so $\ker \tau_H \neq H$, the trivial subgroup. Now,

$$G/\ker \tau_H \cong G \cong \text{Im } \tau_H \leq S_{G/H} \cong S_3$$

and since G and S_3 both are order 6, thus $G \cong S_3$ $\square$

Thm. Let G be a finite group and let $\pi: G \rightarrow S_G$ be the left regular representation. Given $x \in G$, the permutation $\pi(x)$ is a product of $|G|/|x|$ numbers $|x|$ -cycles.

Conjugation action.

Def.

Let G acts G itself by Conjugation:

$$G \times G \rightarrow G : (g, x) \mapsto gxg^{-1}.$$

With an affored Homomorphism

$$\pi : G \rightarrow S_G : g \mapsto (x \mapsto gxg^{-1})$$

Observe that:

$$\ker \pi = \{g \in G \mid gxg^{-1} = x \text{ for all } x \in G\} = Z(G)$$

$$\operatorname{Im} \pi = \operatorname{Inn}(G) \leq \operatorname{Aut}(G) \leq S_G.$$

By the first-isomorphism theorem,

$$G/Z(G) = G/\ker \pi \cong \operatorname{Im} \pi = \operatorname{Inn}(G)$$

And, the stabilizer of $x \in G$

$$G_x = \{g \in G \mid gxg^{-1} = x\} = C_G(x).$$

and the orbit

$$\operatorname{Orb}_G(x) = \{gxg^{-1} \mid g \in G\}.$$

Prop. Let G be a group. Then,

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(x_i)|$$

Where $x_1, \dots, x_r$ are representatives of conjugacy classes not contained in the center,

Conjugate on subgroups.

Let G acts on $\{H \leq G\}$ by conjugation

$$G \times \{H \leq G\} \rightarrow \{H \leq G\} : (g, H) \mapsto gHg^{-1}.$$

With the afforded Homomorphism

$$\pi_G: G \rightarrow S_{\{H \leq G\}} : g \mapsto (H \mapsto gHg^{-1})$$

Observe that: $\text{Stab}(H) = \{g \in G \mid gHg^{-1} = H\} = N_G(H)$.

Directly, $\#\text{Orb}_G(H) = \#\{gHg^{-1} \mid g \in G\} = |G|/|N_G(H)|$.

In particular, when G is finite group of order $p^n \cdot m$, there is order p^n subgroup $P \leq G$,

$$\text{Orb}_G(P) = \text{Syl}_p(G) = \{P, g_2Pg_2^{-1}, \dots, g_{n_p}Pg_{n_p}^{-1}\}.$$

Conjugate on a normal subgroup.

Let G acts on a normal subgroup $H \trianglelefteq G$ by conjugation

$$G \times H \rightarrow H : (g, h) \mapsto ghg^{-1} \quad (\text{It is well-defined by the normality.})$$

With the afforded Homomorphism

$$\pi_H: G \rightarrow S_H : g \mapsto (h \mapsto ghg^{-1}).$$

Observe that: $\ker \pi_H = C_G(H)$ and $\text{Im}(\pi_H) \leq \text{Aut}(H)$.

Prop. Given a subgroup $H \leq G$,

$$N_G(H)/C_G(H) \cong \text{Im} \pi_H \leq \text{Aut}(H).$$

In particular, $G/Z(G) \hookrightarrow \text{Aut}(G)$.

Summary.

Let G be a group, and $H < G$ be a subgroup, $N \triangleleft G$ be a normal subgroup.

Left Coset Action. G acts on G/H by $(g, xH) \mapsto gxH$.

Afforded Homomorphism: $\pi_H: G \rightarrow S_{G/H}: g \mapsto (xH \mapsto gxH)$.

Kernel: $\ker \pi_H = \text{Core}_G(H) = \bigcap_{x \in G} xHx^{-1}$

Stabilizer: $\text{stab}(xH) = xHx^{-1}$.

Orbit: $\text{orb}(H) = G/H$, transitive.

Conjugate itself. G acts on G by $(g, h) \mapsto ghg^{-1}$.

Afforded Homomorphism: $\varphi: G \rightarrow S_G: g \mapsto (h \mapsto ghg^{-1})$

Kernel: $\ker \varphi = Z(G)$

Image: $\text{Im } \varphi = \text{Inn}(G) \leq \text{Aut}(G) \leq S_G$

Stabilizer: $\text{stab}(x) = C_G(x)$

Orbit: $\text{orb}(x) = C(x) = \{gxg^{-1} \mid g \in G\}$.

Conjugate on subgroups. G acts on $\{K \leq G\}$ by $(g, K) \mapsto gKg^{-1}$.

Afforded Homomorphism: $\psi: G \rightarrow S_{\{K \leq G\}}: g \mapsto (K \mapsto gKg^{-1})$.

Kernel:

Image:

Stabilizer: $\text{stab}(K) = N_G(K)$

Orbit: $\text{orb}(K) = \{gKg^{-1} \mid g \in G\}$

[prop. (Dummit, sec 4.3.1a).

Let G be a group and $H \trianglelefteq G$ be a normal subgroup.

Let K be a conjugacy class of G contained in H , and let $x \in K$.

Then, K is a union of $|G|/|C_G(x)|$ conjugacy classes of equal size in H . That is,

$$|C_G(x)| = \bigcup_{i=1}^K |C_H(x_i)| \text{ where } K = |G|/|C_G(x)|, |C_H(x_i)| = \frac{|C_G(x)|}{|G|/|C_G(x)|}.$$

Proof. Notice that since H is normal, for any $x \in H$, $C_G(x) \subseteq H$.

Let G act on G itself by conjugation and consider the afforded homomorphism

$$\pi: G \rightarrow S_G: g \mapsto (h \mapsto ghg^{-1})$$

Then, $\ker \pi = \{g \in G: ghg^{-1} = h \ \forall h \in G\} = Z(G)$, and $\text{Orb}_G(x) = C_G(x)$, and $\text{Stab}_G(x) = C_G(x)$. Therefore, $|G|/|C_G(x)| = |C_G(x)|$.

Meanwhile, similarly, consider the afforded homomorphism

$$\pi_H: H \rightarrow S_H: h \mapsto (k \mapsto hkh^{-1}).$$

Since $C_H(x) = H \cap C_G(x)$ by just definition,

$$|C_H(x)| = \frac{|H| \cdot |C_G(x)|}{|H \cap C_G(x)|} = \frac{|H| \cdot |C_G(x)|}{|C_H(x)|}$$

Application to finite group analysis.

Prop. Every group of order pq where p and q are distinct prime with $p \nmid q-1$ is abelian

